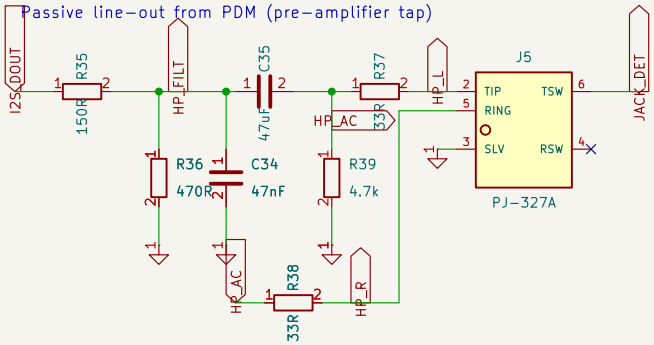
AUDIO OUTPUT

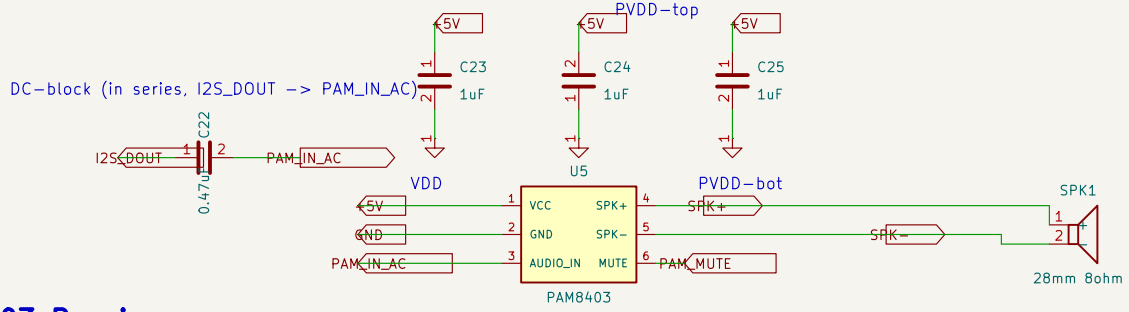
I2S DAC -> PAM8403 Class-D Amplifier -> 28mm 8ohm Speaker

I2S Bus (PDM TX mode - DOUT only):

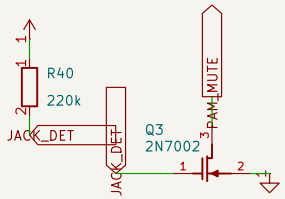
I2S_BCLK	GPIO15 - Bit Clock (UNUSED - PDM mode, v2 free)
I2S_LRCK	GPIO16 - L/R Word Select (UNUSED - PDM mode, v2 free)
I2S_DOUT	GPIO17 - PDM sigma-delta out @ 32kHz -> C22 -> PAM_IN_AC

HEADPHONE OUT + AUTO-MUTE



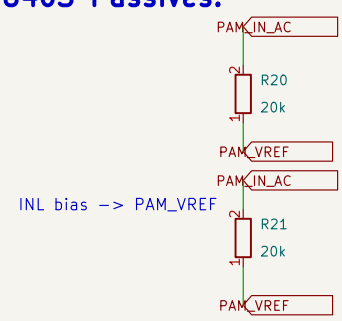


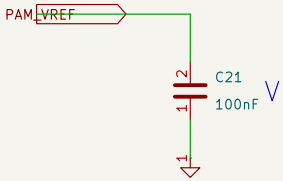
28mm speaker
8 ohm / 0.5W



Plug inserted -> TSW opens -> R40 pulls JACK_DET high -> Q3 pulls PAM8403 MUTE low (speaker off). Unplugged: TSW ties the gate to the tip's ~0V DC (audio peaks < 2N7002 Vgs(th) min - no chatter).

PAM8403 Passives:





VREF bypass (pin 8)

INR bias -> PAM_VREF
R20/R21 bias the amplifier inputs to PAM_VREF (R4-HIGH-3).

Design Notes:

- PAM8403: filterless Class-D stereo amplifier (3W per channel max)
- Only one channel used for mono audio
- No external DAC needed: ESP32-S3 has built-in I2S with DMA
- Powered from +5V rail for max headroom (3.3V limits volume)
- DMA-driven audio streaming for low CPU overhead

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ESP32 Emu Turbo - Handheld Retro Console

Sheet: /
File: 04-audio.kicad_sch

Title: Audio - I2S -> PAM8403 -> Speaker + Headphone Jack

Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.5		Id: 4/1